

Waitlist COLA: A Record-Blind NBA Draft Order With No Tanking Reward

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

Introduction

National Basketball Association (NBA) teams that fall out of the playoff race often stop trying to win, because finishing worse earns a better draft pick. Roughly a third of the league's teams were openly doing this in the 2025-26 season. The league has spent ten years adjusting its lottery odds in response, most recently in May 2026 with a sixteen-team format called 3-2-1. Banchio and Munro (SSAC 2020) proved that odds alone cannot fix it, because any rule giving worse records better odds still rewards losing.

Carry-Over Lottery Allocation (COLA) reads something else. Each team carries a waiting number that rises every year it misses the playoffs and falls when it succeeds, and draft priority follows that number, so one bad season barely moves a team. COLA is a family of rules, and no version had been tested against the others. We tested them, and found the reward for losing in a part of the draft nobody argues about.

Methods

We describe the family with seven settings and run nine draft rules through ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move. Each rule ran 48 separate leagues of 15 seasons, 14,004 league-seasons in all. Every rule answers one question asked two ways. How many places earlier does the first-ranked team draft than the fifth?

Six of the nine rules are COLA versions. One is new. Waitlist COLA keeps the four-pick televised draw exactly as the original runs it, and changes only what comes after. The original fills picks 5 through 14 in order of record; Waitlist COLA fills them by waiting number, longest wait first, ties broken at random.

What the rule reads	Reward for losing <i>draft places</i>	Vs. 2026 rule
Both parts read record (rule through 2026)	1.30	baseline
Draw ignores record, later picks by record	1.98	1.52×
Later picks ignore record, draw unchanged	0.61	0.47×
Neither part reads record	−0.16	noise floor

Table 1. Where the reward for losing comes from. Each number is how many places earlier the worst team drafts than the fifth-worst, on average, which is what finishing last buys. Through 2026 the lottery drew four picks and handed out the rest in order of record. Each row below the first makes one part ignore record. The two parts work against each other, so the ratios do not add up. A weighted draw pulls the worst teams out of the record-ordered picks, hiding how much those picks are worth.

Results

The reward for losing came mostly from those ten picks, which went in order of who lost most. Flattening the odds and leaving them alone makes tanking pay better. The league's new rule confirms it. 3-2-1 draws all sixteen places, so no pick goes by record, and that alone carries its whole effect. Its new odds would have paid tanking more than the old rule.

Several COLA versions remove the reward. Waitlist COLA does it while leaving the lottery fans watch untouched, which the steeper versions give up. Dropping a team from fifth-worst to worst gains 1.87 places under the original COLA and 1.32 under the 2026 rule, and nothing under Waitlist COLA across 720 seasons. The pattern repeats on 26 seasons of real NBA results.

Mechanism	Ranked by record <i>draft places; want 0</i>	Ranked by own standard <i>draft places; want wide</i>
NBA lottery, 2019 rules	+1.09 ± 0.12	+1.09 ± 0.12
3-2-1 as adopted	−1.22 ± 0.25	−1.22 ± 0.25
Classic COLA (original)	+1.88 ± 0.11	+1.38 ± 0.24
Waitlist COLA	−0.12 ± 0.22	+1.71 ± 0.13

Table 2. The same measurement under two rankings. Rank the fourteen lottery teams, take the first and the fifth, and compare the picks they get. Ranked by record, the number is what losing buys, and zero is the goal. Ranked by the rule's own standard of who deserves help, a big positive number means the rule delivers. When a rule's standard is record, both rankings give the same list, so the two columns must match. A negative number flips the reading, so 3-2-1 gives its worst teams the latest picks.

Conclusion

Ending the reward for losing takes no new lottery. Keep the draw, order the picks behind it by waiting number, and a team gains nothing from losing while the longest-suffering teams still pick earliest, more than the rule the league just adopted delivers. Lottery night would look the same to fans. That format expires after the 2029 draft, and the seven settings map the rest of the space.